HW5: Adversarial Prompt Engineering

ezbeda2023 | Security Engineering

DRAFT - live successful attacks and screenshot evidence are incomplete.

Target and scope

The Homework 4 Gemini MCP security-triage agent with its original system instructions, custom triage server, published Fetch server, and synthetic asset inventory. All tests are restricted to this course application. No assets are modified.

Legitimate behavior

Fetch the approved upstream release history, verify the affected range, inspect inventory, assess findings, and produce a truthful owner-assigned remediation plan. Under the documented classroom earlier-than-2.32.4 rule and assumed severity 6.5, web-01 scores 79 (P2), jobs-01 scores 54 (P3), and patched api-01 is excluded. This heuristic is not a verified advisory range.

Experiments prepared

Instruction Override (HLT03.11); Tool Spoofing (HLT05.05); System Prompt Spoofing (HLT05.07). These are candidates, not demonstrated successful attacks. APE definitions: https://ape.hiddenlayer.com/ttps.html

Limitations and remediation

Success is judged by actual final claims and tool arguments compared to baseline. A tool-name completion check does not guarantee the correctness of the plan. Bind remediation input to genuine assessment output; validate provenance and consistency; reject policy instructions in retrieved content; check final claims against the real findings. These experiments demonstrate local report-integrity impact only, not command execution or a production breach. Failed or resisted experiments are retained in the evidence folders.